5. VELOCITY OF SOUND IN AIR USING RESONANCE COLUMN

AIM

To determine the velocity of sound in air at room temperature using the resonance phenomenon.

APPARATUS REQUIRED

Resonance tube, three tuning forks of known frequencies, a rubber hammer, one thermometer, plumb line, set squares, water in a beaker.

FORMULA

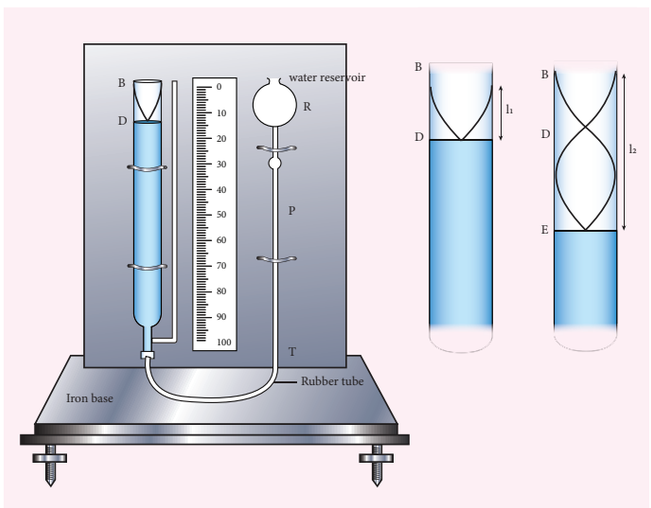
$$V = 2v\left(l_2 - l_1\right) \text{ m s}^{-1}$$

where V ← Speed of sound in air (m s^{-1})

l_1 and l_2 ←The length of the air column for the first and second resonance respectively (m)

v ← Frequency of the tuning fork (Hz)

DIAGRAM



PROCEDURE

- Adjust the position of the resonance tube, so that the length of air column inside the tube is very small.
- Take a tuning fork of known frequency and strike it with a rubber hammer. The tuning fork now produces longitudinal waves with a frequency equal to the natural frequency of the tuning fork.
- Place the vibrating tuning fork horizontally at the open end of the resonance tube. Sound waves pass down the total tube and reflect back at the water surface.
- Length of the water column in the tube is adjusted either by lowering or raising the reservoir or the tube, until a maximum sound(resonance) occurs.
- Measure the length of air column at this position. This is taken as the first resonating length, l_1

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